THE ± 1 CASE FROM A ONE-PARAMETER FAMILY

The argument is DLME's (arXiv:2410.21248, Props. 3.9–3.10 and §5). It is restated here so that the role of each ingredient is visible, and read in the pillowcase language of the uploaded paper.

1. Statement

For a homology sphere Y, $I_\bullet(Y)$ is the $\mathbb{Z}$-graded, Chern–Simons filtered irreducible complex. $\kappa(d)$ is the least filtration level of a nonzero class in degree d, and $\ell(Y) = \inf_d(\kappa(d) - d/8)$. In the normalisation used throughout, this is the least $(3+\rho)/16$ of a nonzero class.

Theorem 1 (DLME)**.** *Let $K \subset S^3$ be nontrivial. Then $\ell(S^3_{-1}(K)) \le \ell(S^3_1(K)) - \frac{1}{8} - \eta(K)$ with $\eta(K) > 0$. In particular $S^3_1(K) \ncong S^3_{-1}(K)$ as oriented manifolds.*

2. Ingredients

(1) The triangle. For $K \subset Y$ there is a distance-two exact triangle (CDX; DLME Prop. 3.9, over $\mathbb{F}_2$)

$$\cdots \to I(Y_1) \xrightarrow{W_*\oplus(W,c)_*} I(Y) \oplus I(Y) \xrightarrow{(W',c')_*\oplus W'_*} I(Y_{-1}) \to \cdots .$$

Pillowcase reading (heuristic: it uses the paper's Conjecture 3.2). L_{-1} is obtained from L_1 by one Dehn twist along each edge $\{a=0\}$ and $\{a=\frac{1}{2}\}$. Both edges are the line of $Y = S^3$: they are $L_{1/0}$ and its χ-translate, which is why $I(Y)$ appears twice. Each twist contributes one Seidel triangle whose third term is the edge. Γ_K misses both edges (Prop. 1.1(b)), so for $Y = S^3$ the third term contributes nothing. This is the distance-two analogue of the uploaded paper's §4.2.

(2) The one-parameter family map. Triangle detection produces a null-homotopy $g_1 \colon I_\bullet(Y_1) \to I_\bullet(Y_{-1})$ of the composite through the middle term. It counts instantons with irreducible ends over a one-parameter family of metrics $G \cong [-1,1]$ on the composite cobordism $W^1_{-1} = W(Y_1 \to S^3) \cup_{S^3} W(S^3 \to Y_{-1})$. At one end the family breaks along the middle S^3. At the other it breaks along a hypersurface with $\mathbb{R}P^3$ components (DLME §5). The $\mathbb{R}P^3$ bounds the (-2)-sphere formed by the cocore of the first handle and the core of the second. When the middle term is 0, as for $Y = S^3$, g_1 is a chain map and a quasi-isomorphism (DLME Prop. 3.10, §5).

(3) Its level. W^1_{-1} has $b_1 = b^+ = 0$ and $\pi_1 = 1$: each piece is a 2-handle attached to $S^3 \times I$. An instanton counted over a d-parameter family has index $i_0 = -d$. The level formula $c(\beta) - c(\alpha) = \frac{i_0 + 3b^+}{8} - E$ then gives

$$c(\beta) - c(\alpha) = -\tfrac{1}{8} - E.$$

The bundle drops out. In DLME's terms, with $c^2 = -1$: degree $D = 1 - 2c^2 = 3$, level $L = -c^2/4 - E$, and $L - D/8 = -\frac{1}{8} - E$.

(4) Positivity. $E > 0$, because a flat connection on the simply connected W^1_{-1} is trivial and cannot have irreducible limits. Compactness gives finitely many counted instantons, so $\eta(K) := \min E > 0$.

(5) Nonvanishing. $I(S^3_{\pm 1}(K)) \cong I^w(S^3_0(K)) \neq 0$ by Floer's triangle and Kronheimer–Mrowka. Hence ℓ is finite.

Lemma 2 (DLME Lemma 2.7)**.** *Let f be a filtered chain map of degree D and level L that is injective on homology in every degree. Then $\ell(\text{target}) \le \ell(\text{source}) + L - D/8$.*

Proof. A class at level $\kappa(d) + \epsilon$ maps to a nonzero class at level $\kappa(d) + L + \epsilon$ in degree $d + D$. Subtract $(d+D)/8$ and take infima. □

Proof of Theorem 1. Apply the lemma to g_1, using (2), (3), (4) and (5). □

3. What each ingredient contributes

- *The $\frac{1}{8}$ is the family dimension, and nothing else.* One parameter means one unit of index, which is $\frac{1}{8}$ of level. No ρ-term from a middle end enters the counted maps. The $\mathbb{R}P^3$ appears only at the boundary of the family.
- *Strictness is positivity of energy* on a simply connected cobordism, which is an allowed input.
- *What the cones of the uploaded paper's §6 cannot supply* is the direction of the inequality. Those cones join $Y_{\pm 1}$ through (Y_0, w); that is the horizontal rotation, and it gives only $\ell(Y_1) - \ell(Y_{-1}) = \frac{3}{8} - \text{Area}(T) - \frac{1}{8}\Delta\iota$. For a pair $x \mapsto y$ counted by g_1, Stokes and the level formula (exact for torus knots, plausible in general) give

  $$E = \tfrac{1}{4} - \text{Area}(T_{xy}) - \tfrac{1}{8}\Delta\iota.$$

  So $E > 0$ is precisely the area bound that the cones cannot see. For the right-handed trefoil, $\ell(Y_1) - \ell(Y_{-1}) = \frac{61}{280} = \frac{1}{8} + \frac{13}{140}$.

4. **Why the same proof does not give ± 2**

L_{-2} is obtained from L_2 by *two* twists along each edge. The relevant composite $W' = (-X_2) \cup_{S^3} X_{-2}$ contains a (-4)-sphere bounded by $L(4,1)$. The analogous family map has the same index count, so the same level $-\frac{1}{8} - E$, and $E > 0$ holds again because W' is simply connected. What fails is one of the two structural properties:

- *Bundle even on the sphere:* flat caps on the (-4)-disk bundle with central $L(4,1)$ boundary give uncancelled lens faces. The middle term is not 0, and the family map is not a chain map. This matches the uploaded paper's §4.5 remark that $(\frac{1}{2}, 0)$ is a central flat connection on $Y_{\pm 2}$; the identification is my reading.
- *Bundle odd on the sphere (mixed bundles):* the family map is plausibly a chain map, but it is not injective in general. It would force $m \geq \frac{1}{8}$, whereas $m_2 = \frac{1}{16}$ for $T(2,-3)$: the class is absorbed.

These are DLME's two "inadequate variations". The map that is a quasi-isomorphism, B, has level $+\frac{1}{8}$, because the insertion $\mu(S)$ costs $\frac{1}{4}$. So the ± 2 problem is exactly to find a replacement for g_1 that is injective with level below 0. The ± 1 argument needs nothing beyond the family dimension and positivity because there the middle term vanishes on the nose.